



GOOD MORNING

FAⁿty Radiographs

Faulty Radiographs

Oral Radiology 25

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Learning Objectives

Student should be able to:

- Enumerate different types of radiographic faults
- Describe radiographic density and contrast
- Describe the causes of radiographic faults

DEFINITION

An 'Ideal Radiograph' is the one with optimum density, maximum detail and sufficient contrast to make the detail apparent.

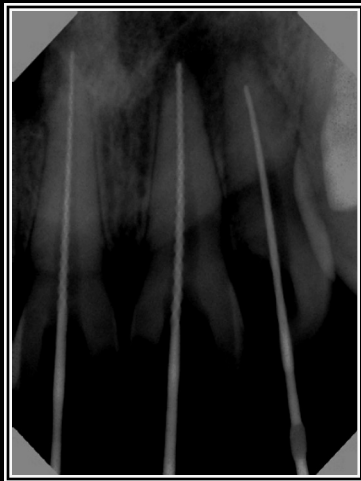
The image should be sharp, centered and have the same size and shape as that of the object.



DENSITY

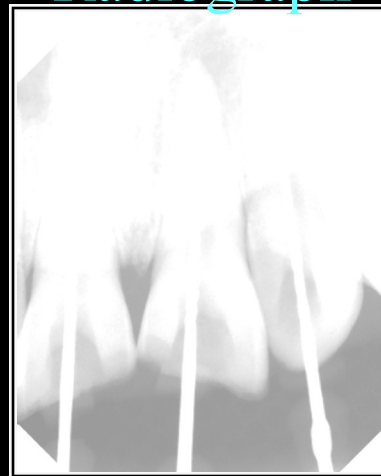
“Overall degree of darkening of a processed film”

Dark
Radiograph



High
Density

Light
Radiograph



Low
Density

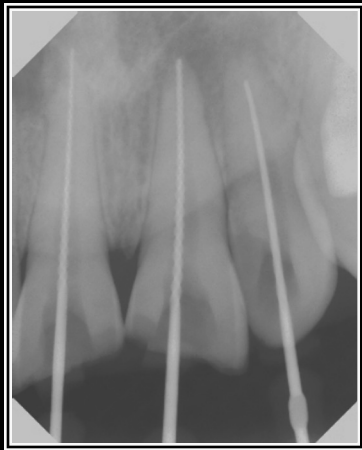
Just Right



Optimum Density

CONTRAST

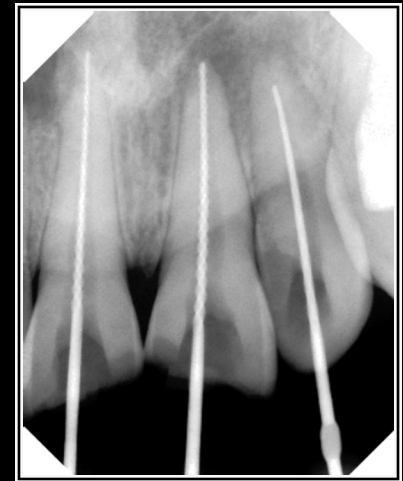
“Difference in densities in different regions of the film”



Low
contrast



High Contrast



Sufficient
Contrast

SHARPNESS (DEFINITION)

“Effectiveness of a radiograph to precisely define an edge”



Blurred Image



Sharp Image

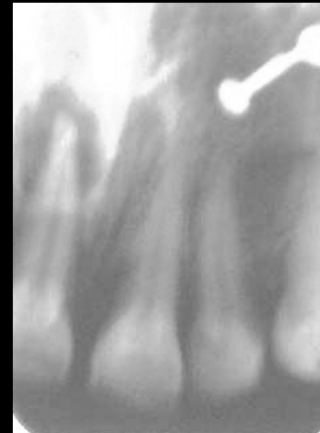
TYPES OF ERRORS

- a) Technique and projection errors*
- b) Exposure and processing/Handling errors*

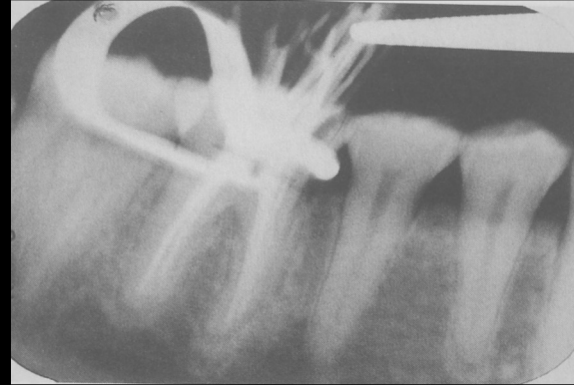
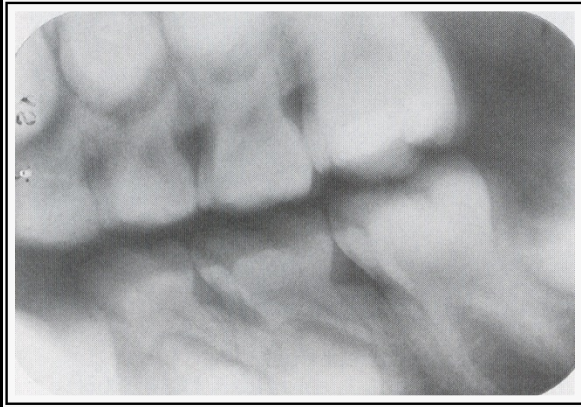
Patient Preparation Errors

- **Radiopaque Artifacts:**

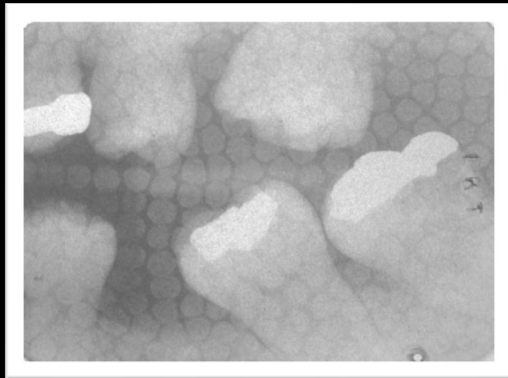
- Dental appliances
- Eye-glasses
- Jewelry
- Wigs
- Bands
- Hair-clips



BLURRED IMAGE



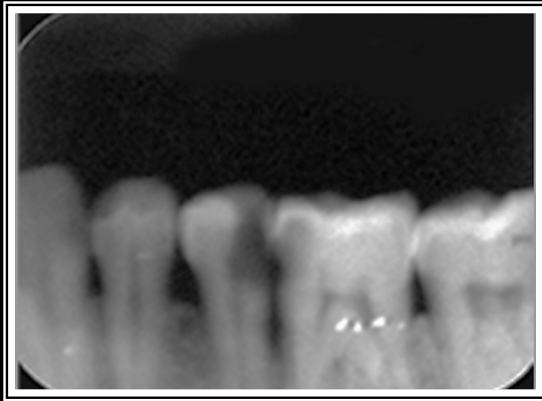
Movement : of patient, tube, film/sensor during exposure.



Backward Film/sensor placed

Film Placement Errors --Partial image

Apical cut-off



Sensor/Film Placement Too Close to teeth (paralleling tech):

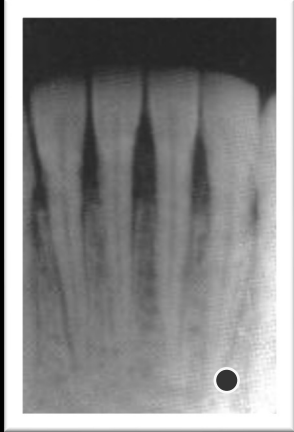
- **Vertical Angulation Too Flat:**
 - Beam not directed steeply enough
- **Sensor Size Too Small:**
 - Does not cover the tooth from crown to apex ($\frac{1}{8}$ th inch)

Crown cut-off



- **Sensor/Film Placement Too Far Back:**
- **Vertical Angulation Too Steep:**
- **Sensor Size Too Small:**
 - Cannot cover the full tooth length

Dot Artifact



Raised dot always occlusally

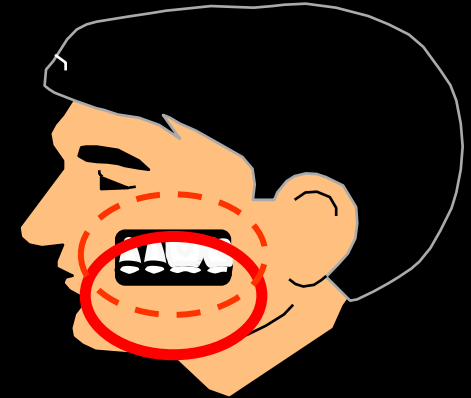
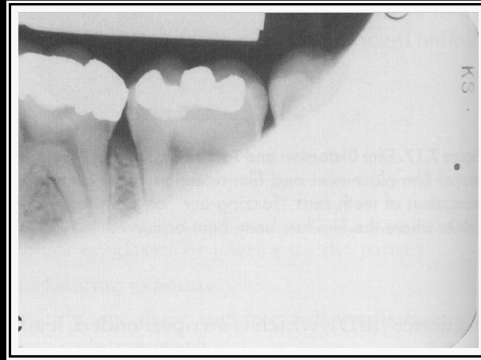
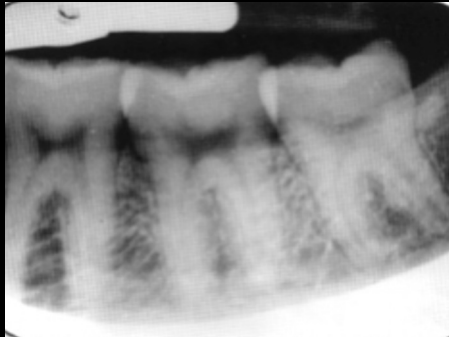
- Using the same Double exposure film/sensor twice
- Poor labeling or handling of PSP plates
- Automatic exposure errors in digital sensors (trigger two exposures due to software or operator error)



FIGURE 4-16 Strange looking teeth...

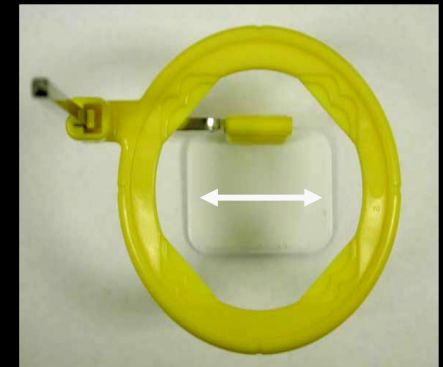


Cone-cut error



PID not properly aligned with film holder

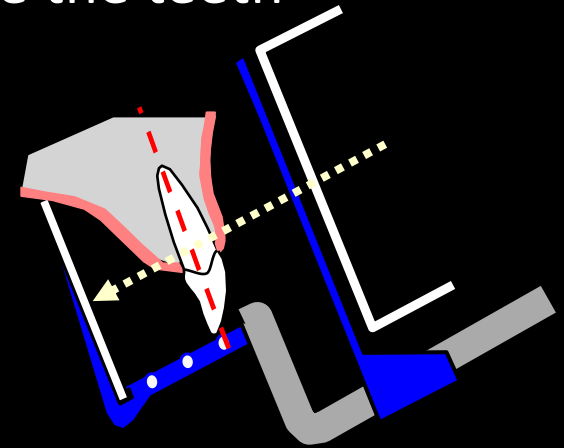
- **Improper alignment of X-ray cone (PID) and sensor:**
- **Incorrect sensor/film placement.**
- **Movement of sensor/film during exposure:**



Shape Distortion Error

Elongation

Elongation is a radiographic distortion where the teeth appear longer than their actual length

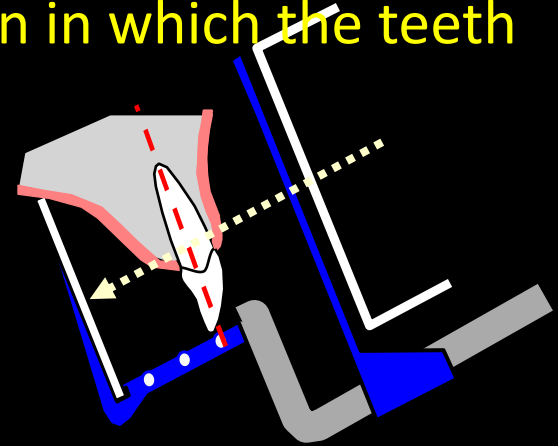


Decreased vertical angulation of X-ray beam:

- **Too flat/low angulation:** The central ray is directed at a **smaller angle** than required relative to the long axis of the tooth
- **Example:** For maxillary teeth, if $+20^\circ$ angulation is needed but less is used, elongation occurs.

Foreshortening

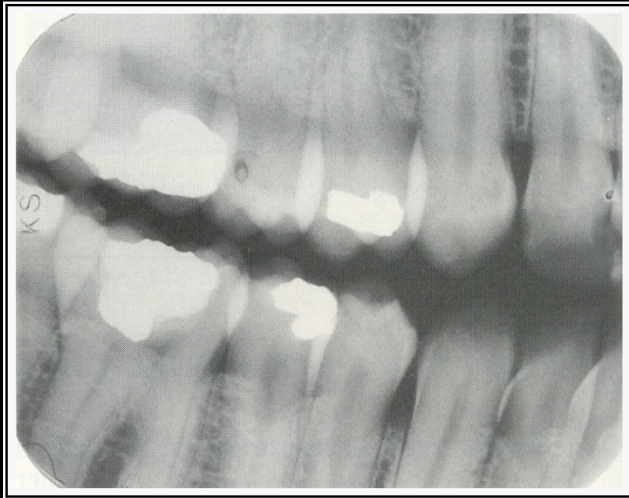
Foreshortening is a radiographic distortion in which the teeth appear **shorter than their actual length**



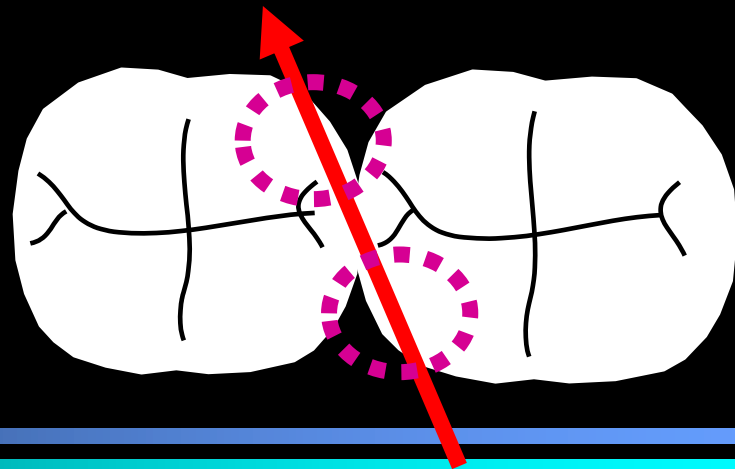
Increased vertical angulation of the X-ray beam:

- The central ray is directed at a **steeper angle** than required to the long axis of the tooth.

Improper Horizontal Angulation



Overlapped image



Film Adaptation : Image distortion

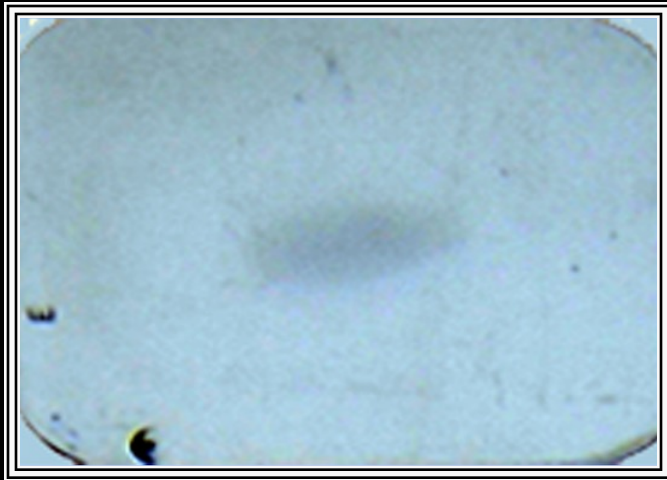


Film bent, pressure.

Unexposed film/sensor

Film-Clear transparent

Sensor- Blank



Failure to switch on x-ray machine

Film not exposed to radiation

Film Adaptation : Image distortion



Film bent, pressure.

Light radiograph-Reduced density



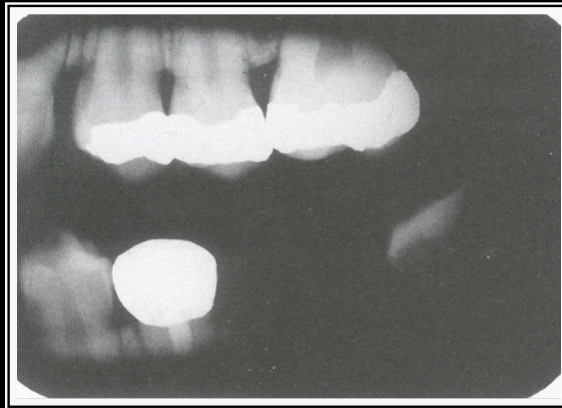
Normal radiograph

- Low exposure time, Low mA, Low kVp
- Increased source-to-film distance
- Thick bone or dense anatomical area
- Expired or improperly stored film

PSP Sensor

- PSP plate not erased before reuse
- Delay in scanning after exposure

Dark radiographs

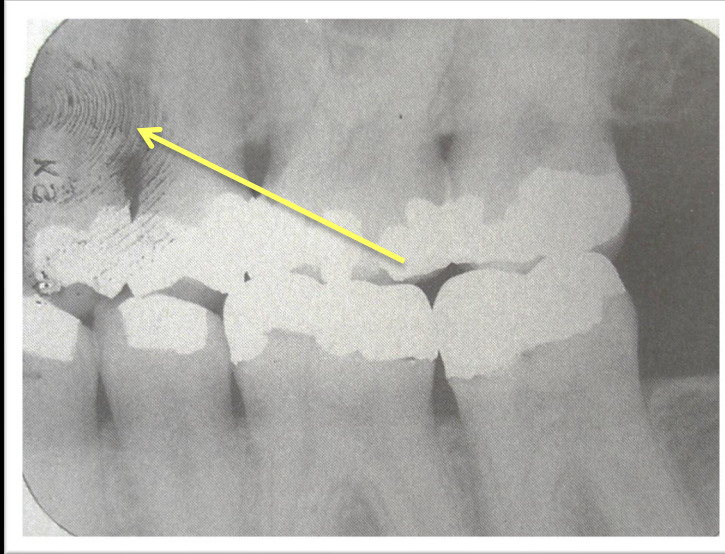


- **Excessive exposure time**
- **High mA**
- **High kVp**
- **Short source-to-film distance**

PSP Sensor-

- **Improper calibration of the PSP scanner**
- **Background radiation exposure before scanning**

Finger Prints



FILM:

Black finger print

Film touched by fluoride or developer contaminated finger.

Handle films by edges only.

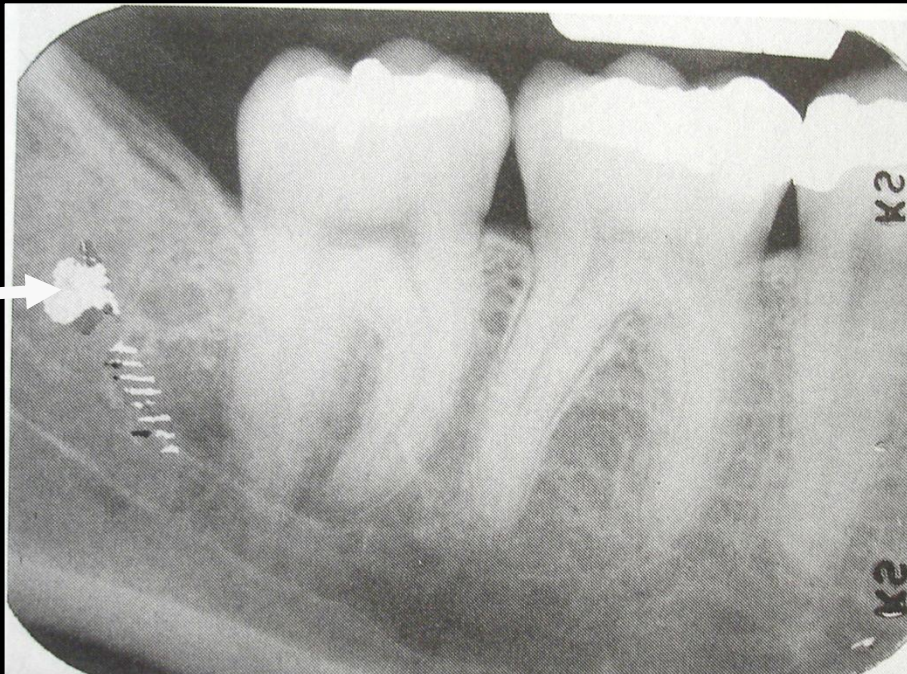
Handling the PSP plate with bare fingers

- Oils, sweat, or moisture on fingers contaminate the plate surface
- These contaminants interfere with **laser scanning**

Radiographic appearance

- Appears as:
 - Dark or light smudges**
 - Curved or whorled patterns**

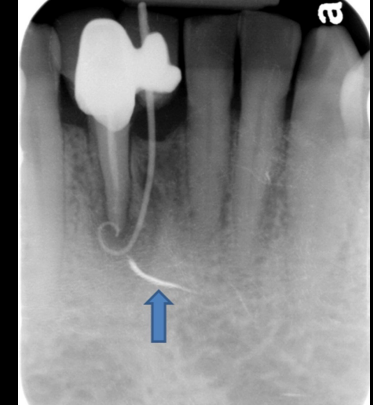
Scratches/emulsion



Improper film handling

Soft emulsion removed from film
by sharp object

Excessive time in wash water



Scratches –PSP

IMPROPER HANDLING

AS –WHITE /DARK
STREAKS

- **Delayed scanning** → loss of latent image (faded / noisy image)
- **Ambient light exposure before scanning** → partial or complete image loss.
- **Scratches on plate** → white or dark lines
- **Dust or debris on plate or scanner** → white spots or lines
- **Scanner malfunction** → streaks or banding artefacts

REFERENCES:

- White And Pharoah's ORAL RADIOLOGY, 8TH Edition, Sanjay M. Mallya, chap 5-pg 215
- Çalışkan, A., & Sumer, A. P. (2017). Definition, classification and retrospective analysis of photostimulable phosphor image artefacts and errors in intraoral dental radiography. Journal of Dental Imaging